5 : Dealing with failures

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Dealing with failures

- Dealing with failures
  - Link failures in large IP networks
  - IP-based fast restoration
  - MPLS-based fast restoration

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The network

- How to measure failures in a network ?
- ISIS data was collected by Abilene on KSCY
- Raw data (December 2004) available from http://abilene.internet2.edu/observatory/

Source : http://abilene.internet2.edu/maps-lists/

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Taxonomy of ISIS events

- Refresh LSP
  - frequency is function of LSP lifetime
- Adjacency down and Adjacency up
  - link up and link down
- Neighbour metric up or down per hour
  - Change in link weight for traffic engineering purposes
- IP prefix down or IP prefix up
  - IP prefix advertised by a router becomes invalid or valid
  - A change in IP prefix status usually follows link change
- IP prefix metric down or up
  - Change in the metric associated to a prefix
- Change in Overload bit
  - Usually set on router reboot during BGP startup
- TE reservation change
  - Change in reserved bandwidth when MPLS-TE is used
- LSP lifetime set to zero

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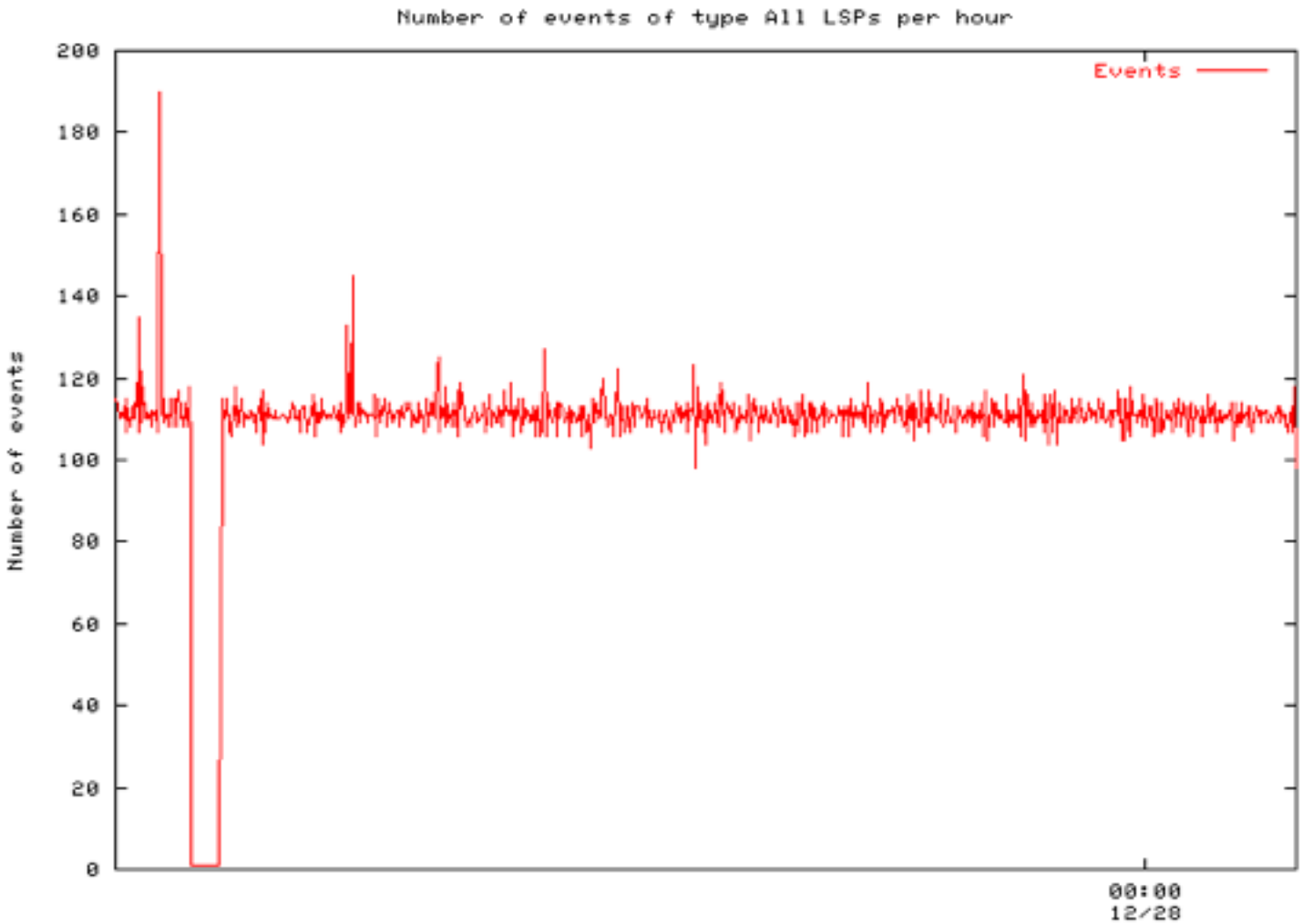
The ISIS LSPs per hour



- Most LSPs are simple refresh

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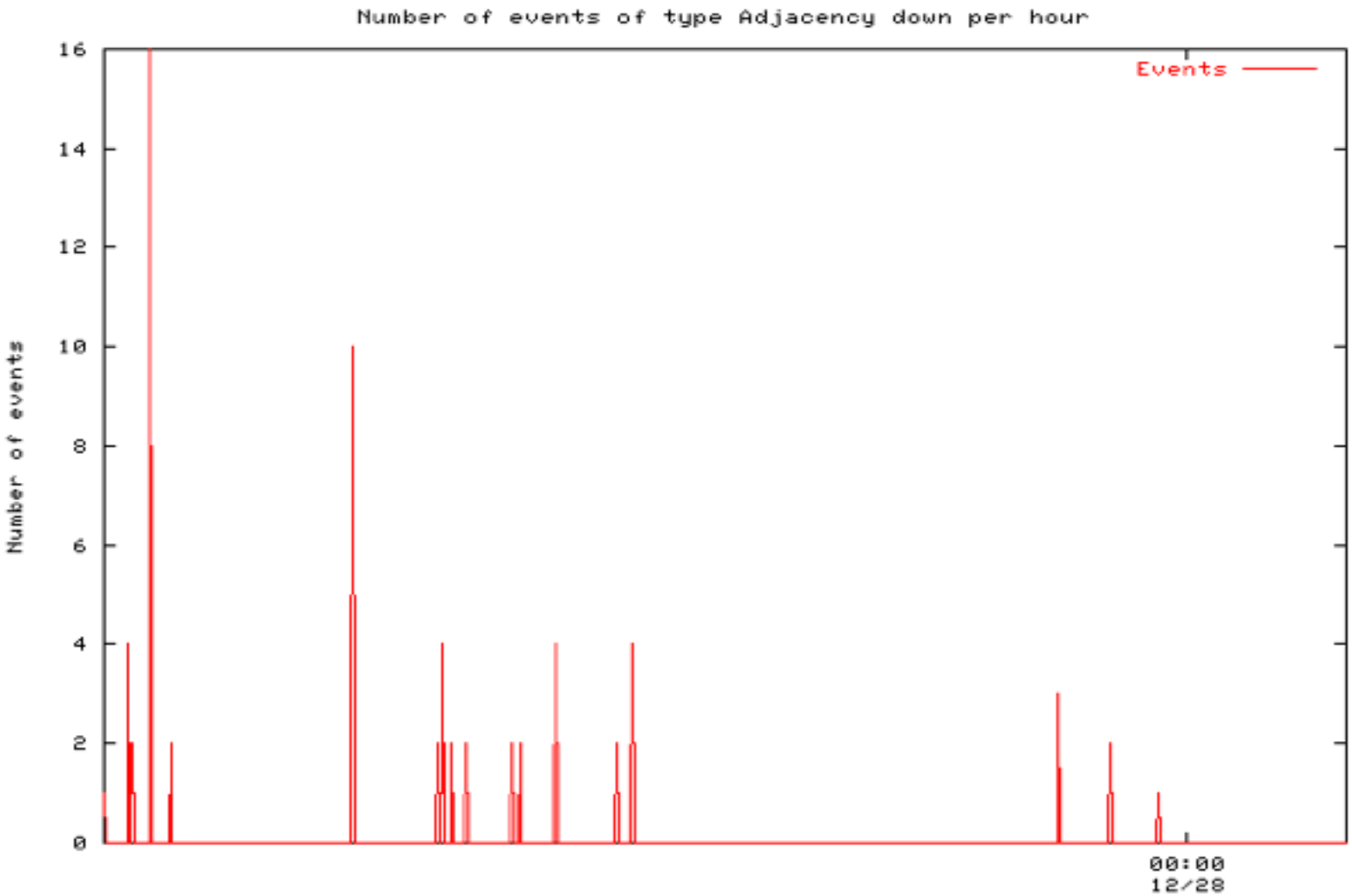
The link down events



- 71 link down events during one month

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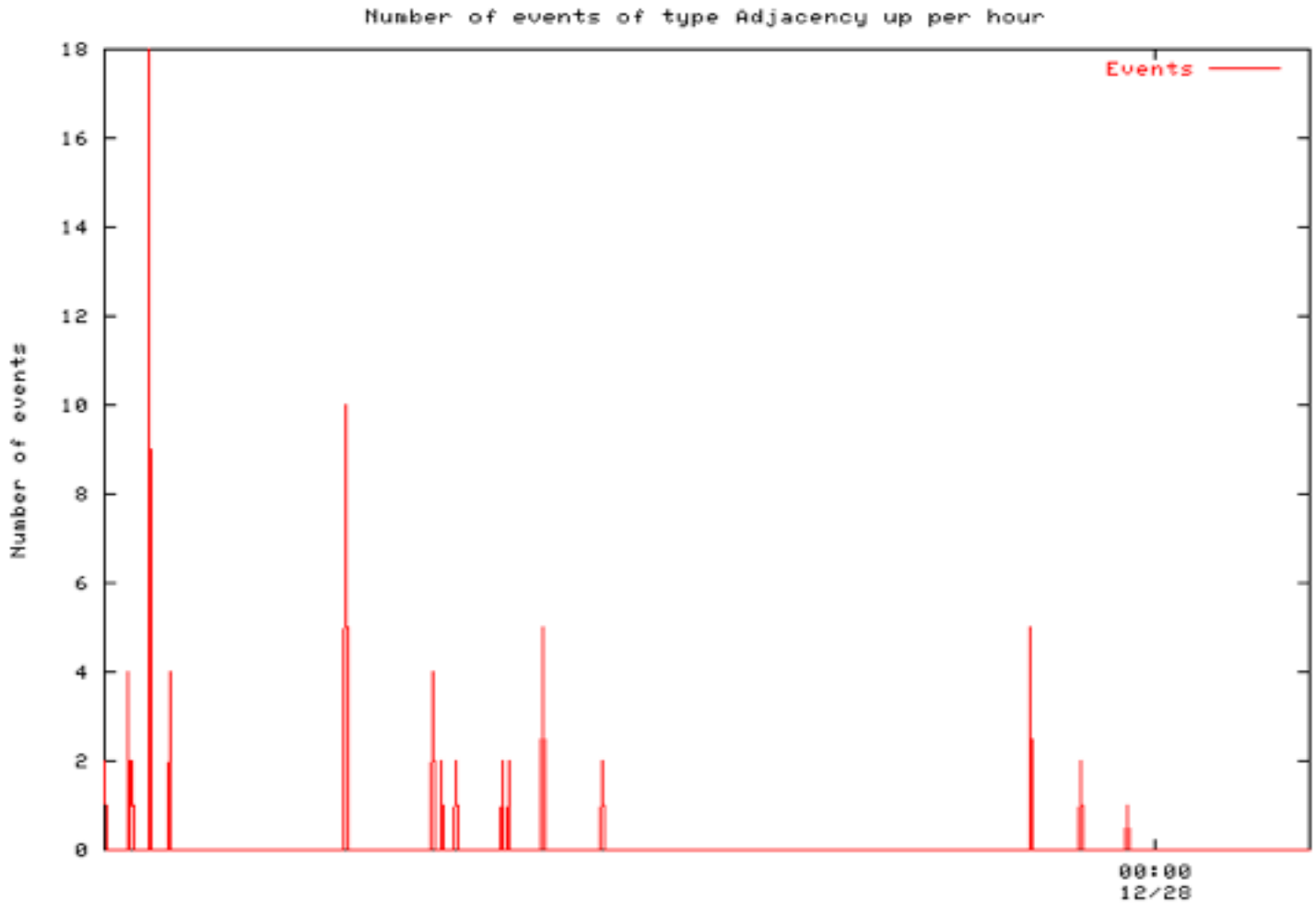
The link up events



- 67 link up events during one month

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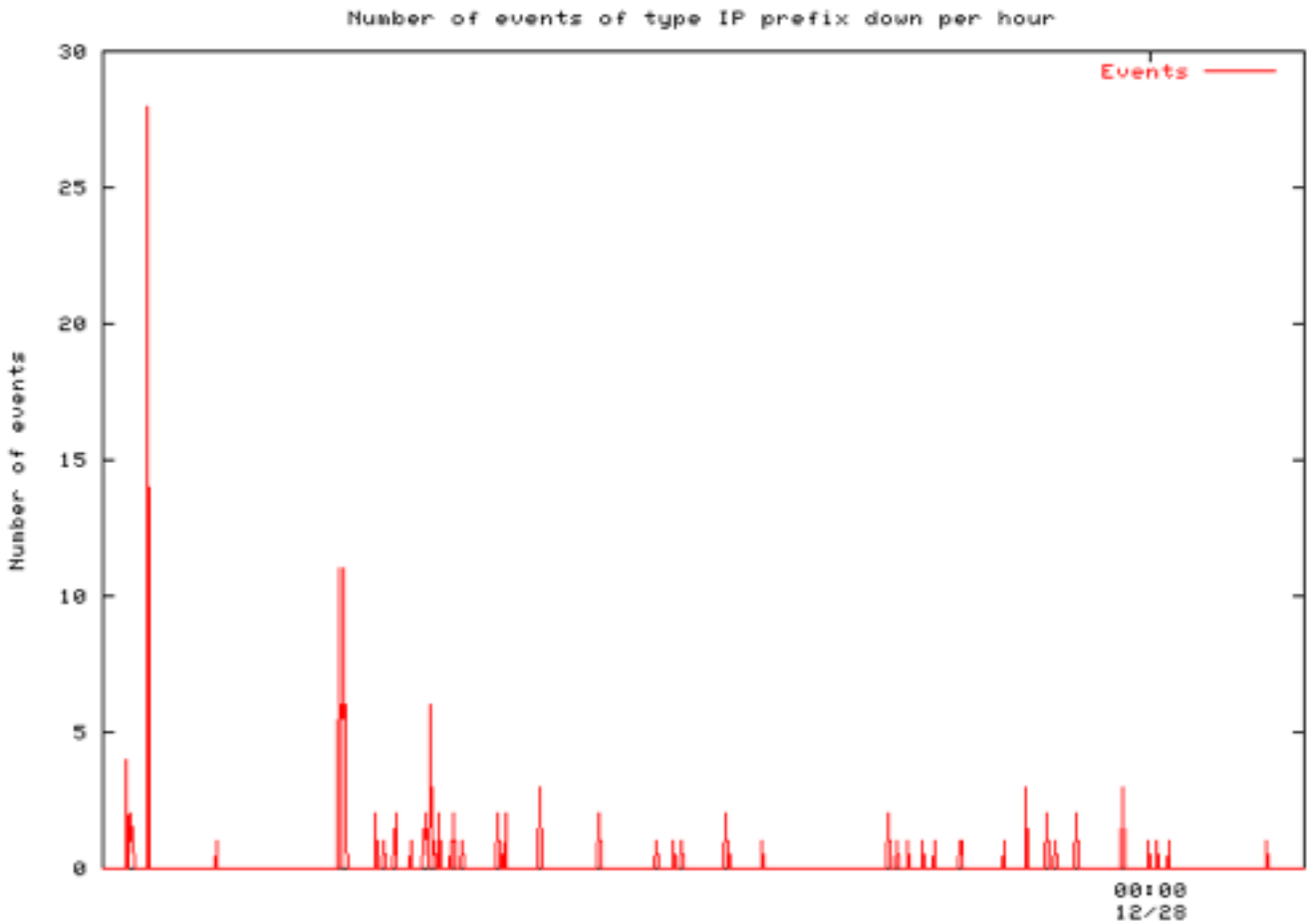
The prefix down events



- 128 prefix down events during one month

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IS-IS in a tier-1 ISP

- The Network
  - Large tier-1 transit ISP
  - 400 routers in studied ISIS area
  - IS-IS wide metrics and TE extensions are used in the network
  - MPLS traffic engineering is enabled
- The trace
  - IS-IS adjacency between a PC running a modified tcpdump and a router
  - all IS-IS packets logged in libpcap format during one month
    - analysed with scripts and lisis
    - http://totem.info.ucl.ac.be/tools.html

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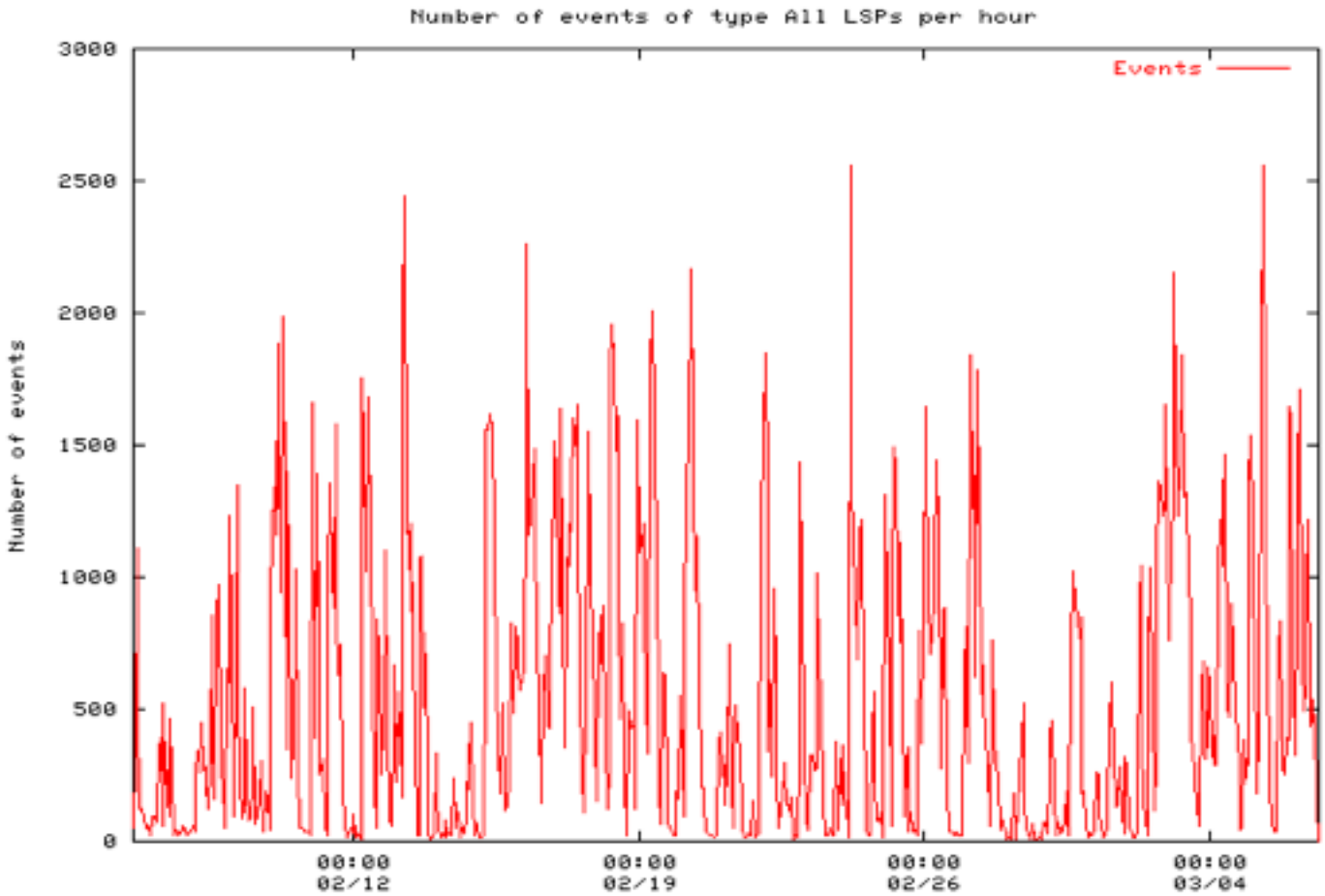
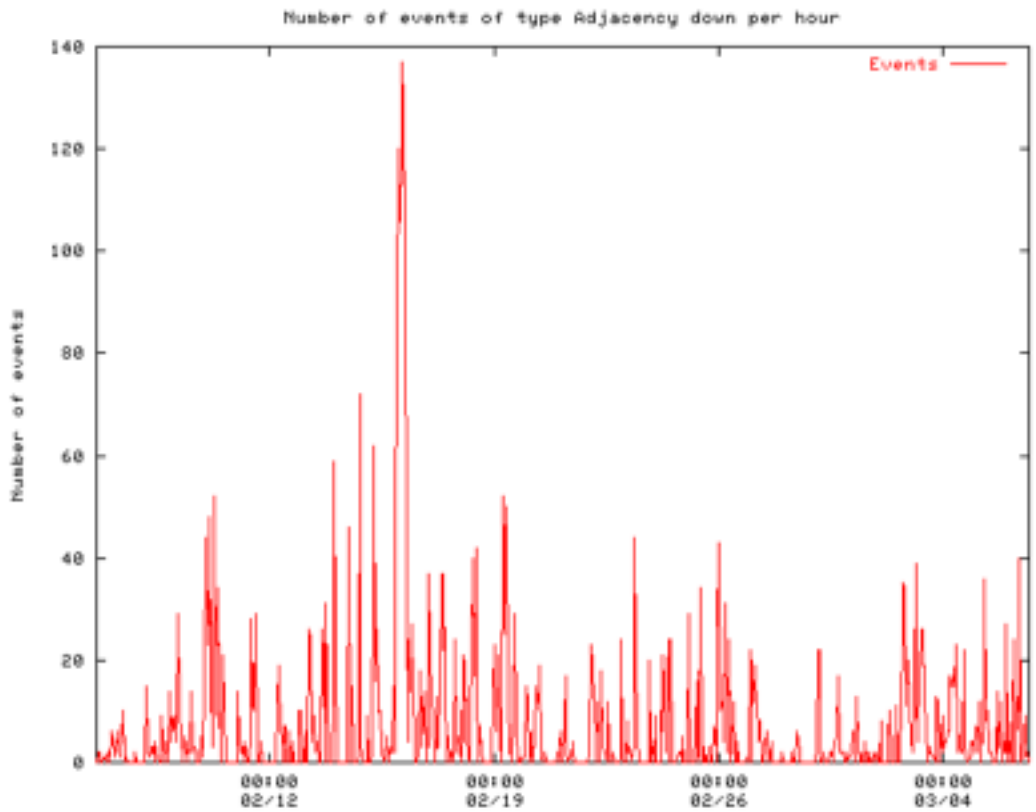
The hourly IS-IS load

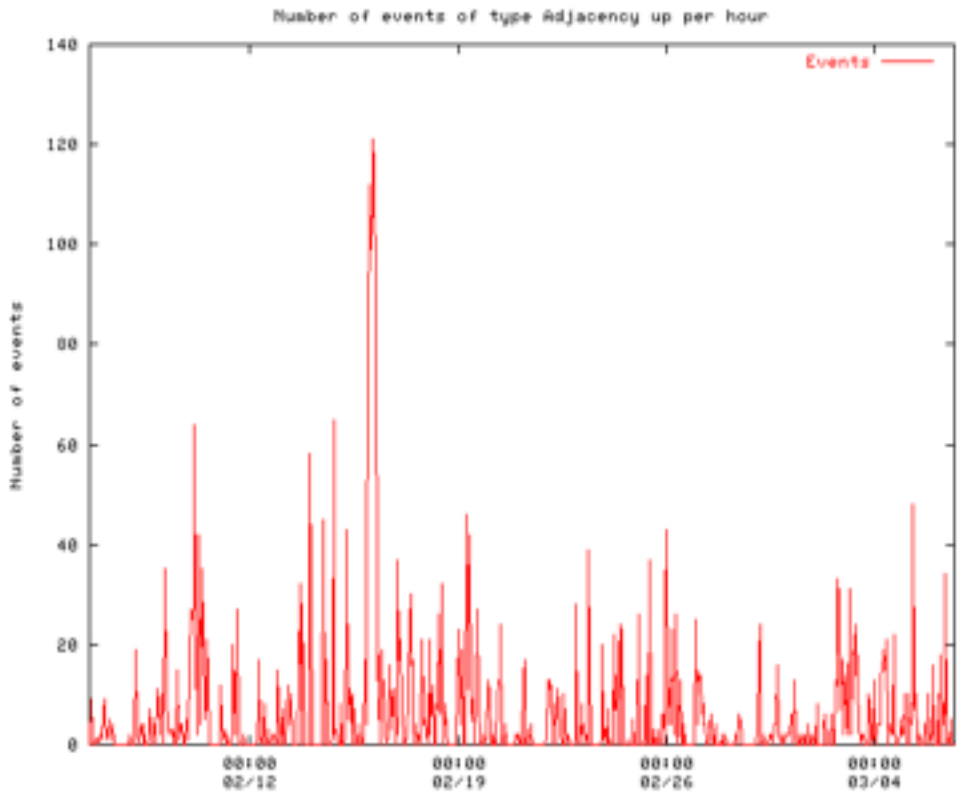
- 367383 collected LSPs during one month
  - up to 2500 LSPs per hour...
  - 6% of those LSPs are refresh LSPs
    - LSP lifetime set to max=65500 seconds



The adjacency changes per hour

- 5276 adjacency down LSPs (left)
  - metric increase events are negligible (40)
- 4487 adjacency up LSPs (right)
  - metric decrease events are negligible





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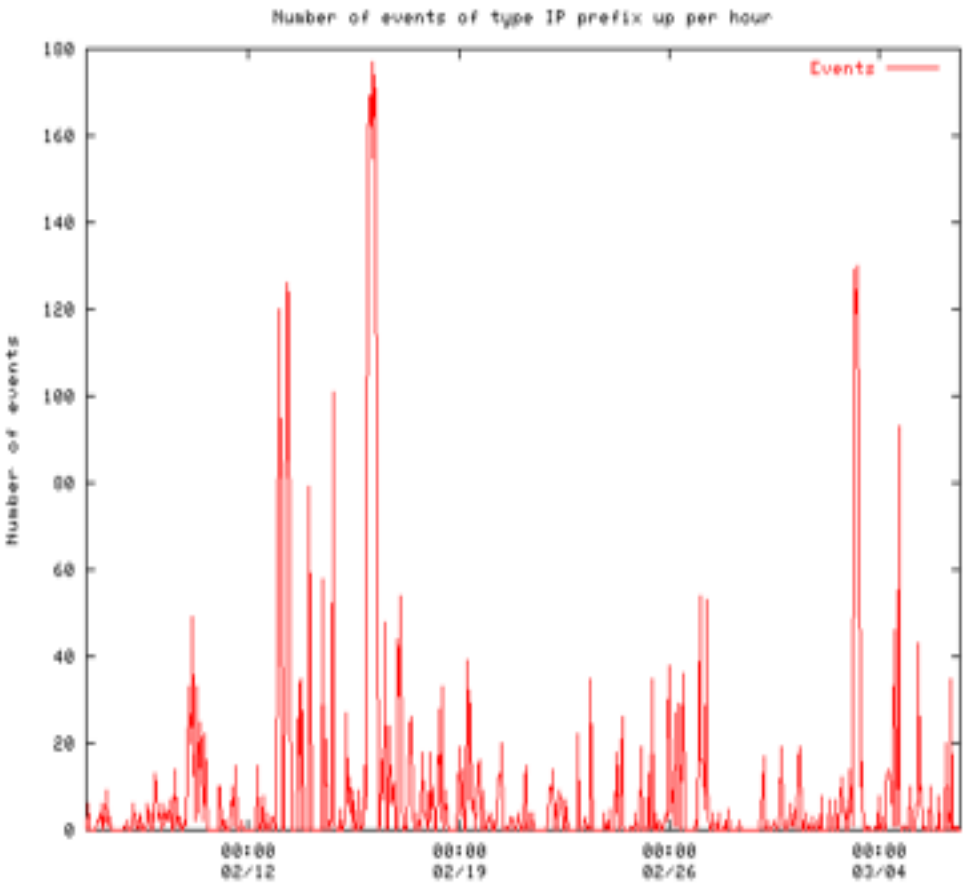
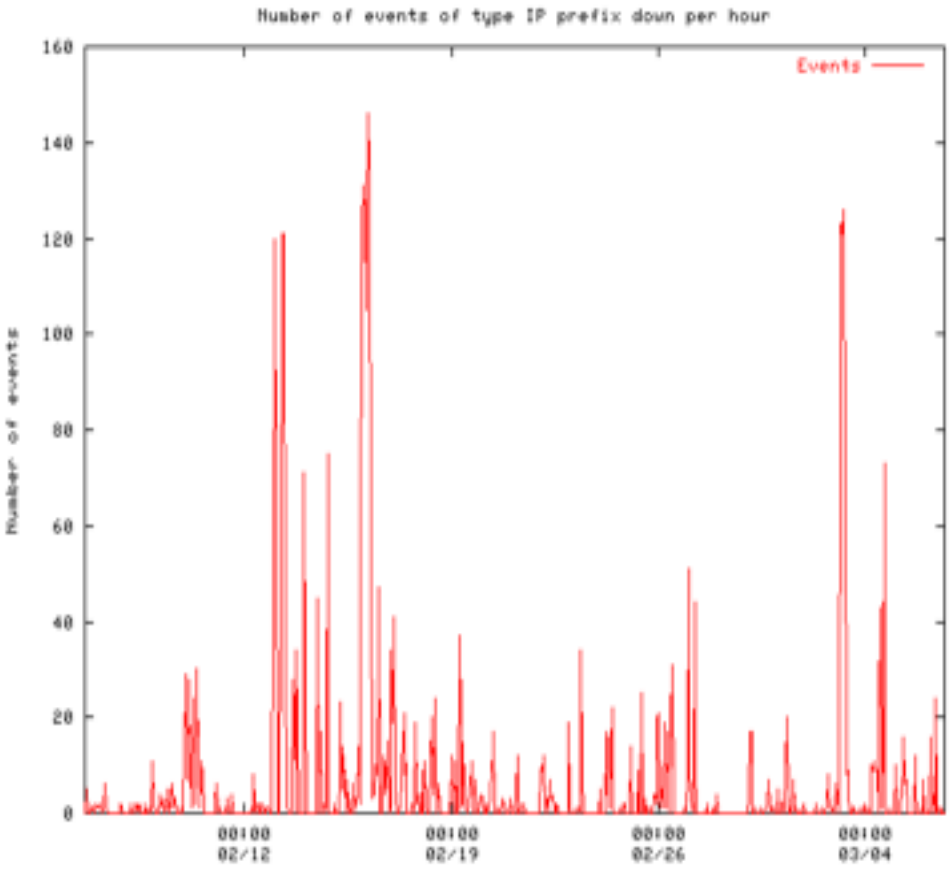
Prefix changes per hour

- Almost no metric changes for prefixes
  Prefix up Prefix down





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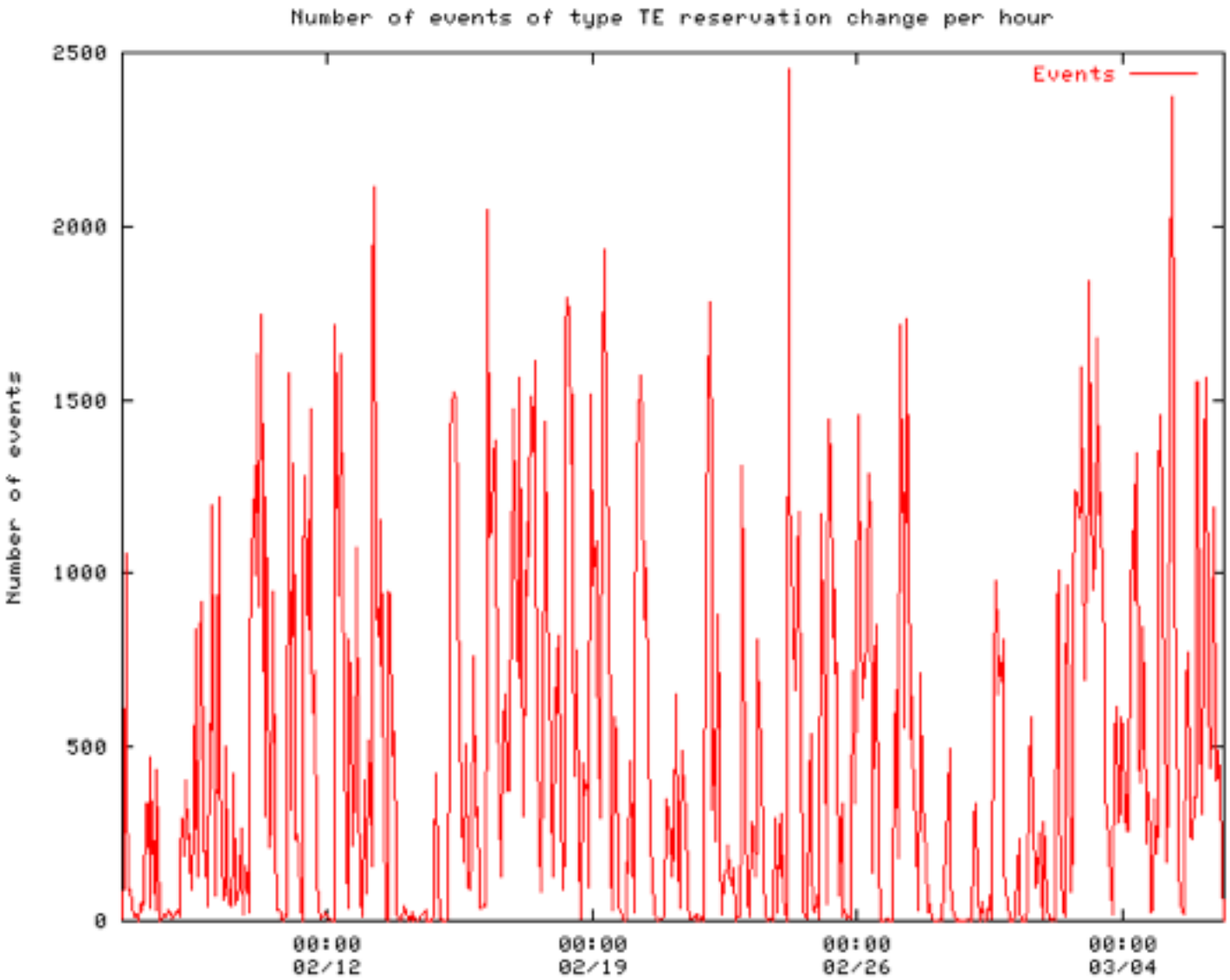
The LSPs with TE changes

- 88% of all LSPs !



Why so many TE LSPs ?

- Routers are supposed to advertise TE info when crossing percentile thresholds :
  - Up : 15 30 45 60 75 80 85 90 95 96 97 98 99 100
  - Down: 100 99 98 97 96 95 90 85 80 75 60 45 30 15
    - Such changes are infrequent
  - Most bandwidth changes are only for 10 Kbps
    - Common value for reserved bandwidth for TE-tunnels with unknown demand
- But, unfortunately those routers also
  - advertise minor TE changes after some delay
    - default value : 3 minutes

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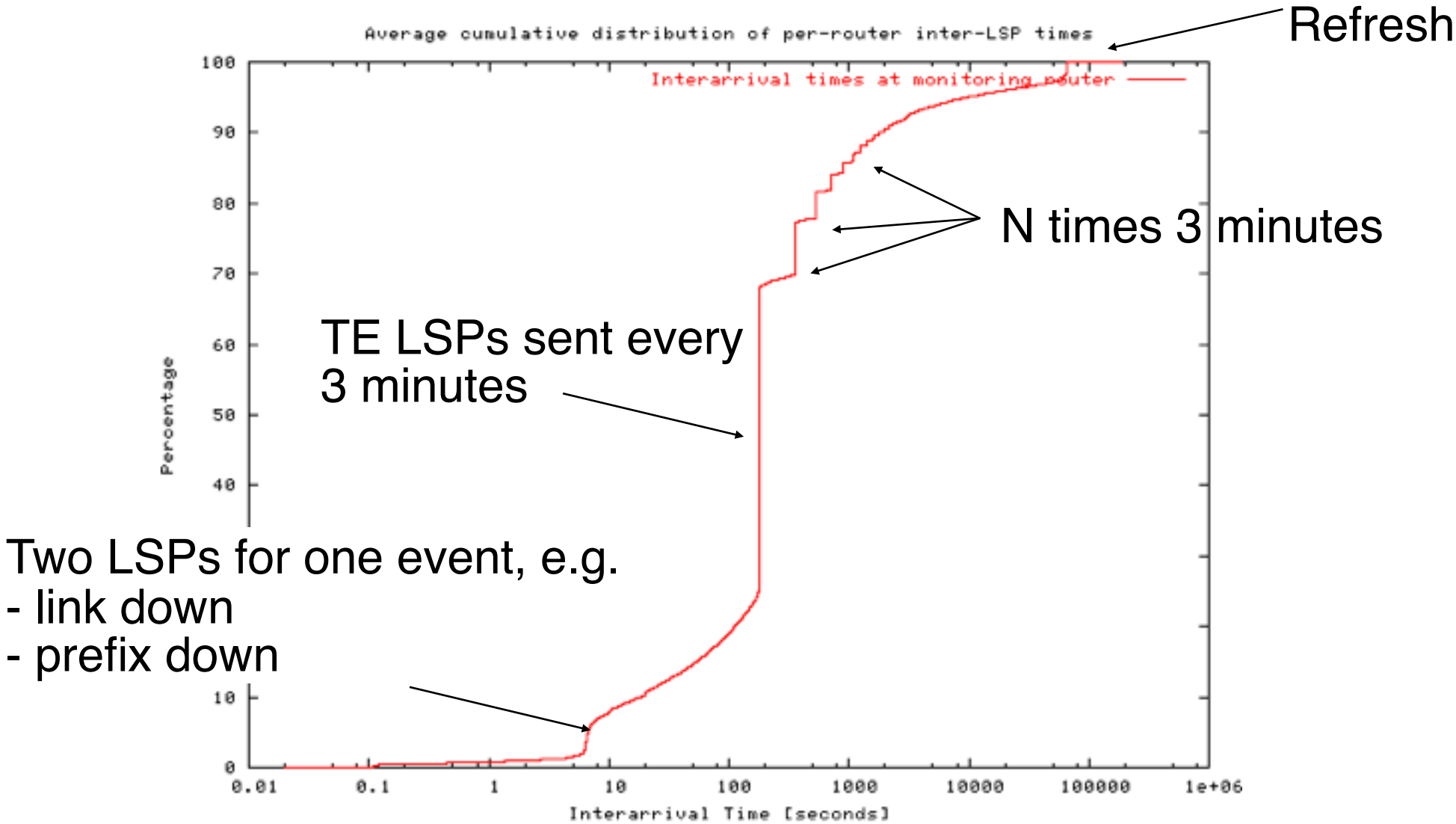
Average per router inter-LSP transmission times



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Dealing with failures

- Dealing with failures
  - Link failures in large IP networks
  - MPLS-based fast restoration
  - IP-based fast restoration

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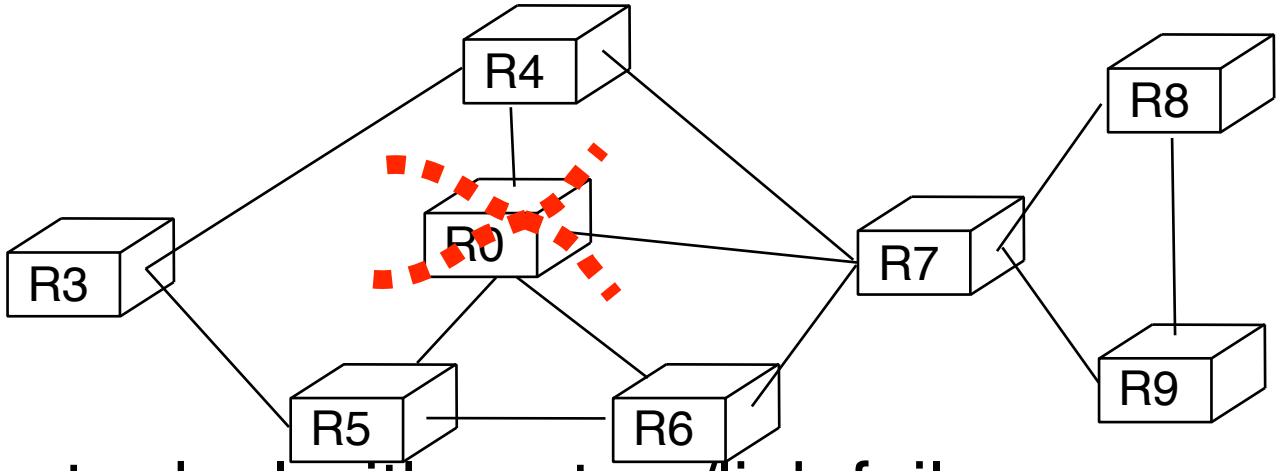
Dealing with link/router failures

- Providing a good service means
  - providing the promised bandwidth/delay ...
    ... even in case of network failures



  - How to deal with routers/link failures
    - Detect the failed component
      - Detection by the routing protocol itself (adjacency information)
      - Detection with the help of layer 2 information (carrier lost)
    - Propagate the bad news inside the network

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Detecting link/router failures

- How to detect link/router failures ?
  - Rely on failure detection of physical layer
    - possible with SONET/SDH and some types of modems
    - detection delay may vary from 10 msec to a few seconds
    - difficult on LAN interfaces such as Ethernet
  - Relying on the routing protocols
    - BGP
      - once peering session has been established, KEEPALIVE messages are sent every 30 seconds
      - peer is dead if no KEEPALIVE within 90 seconds
    - OSPF
      - Hello packets sent every 10 seconds
      - Neighbor is dead if no hello received in 40 seconds
    - RIP
      - router sends routing table every 30 seconds
      - neighbor is dead if no table within the 180 seconds

Untuned routing protocols may need 10s seconds to detect failures !

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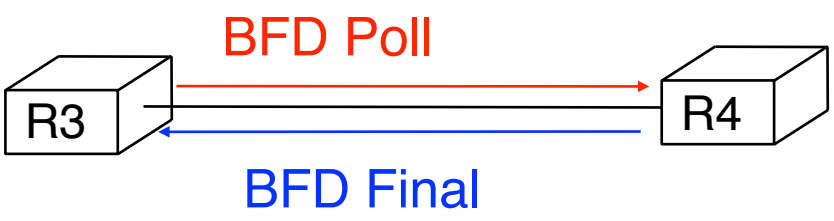
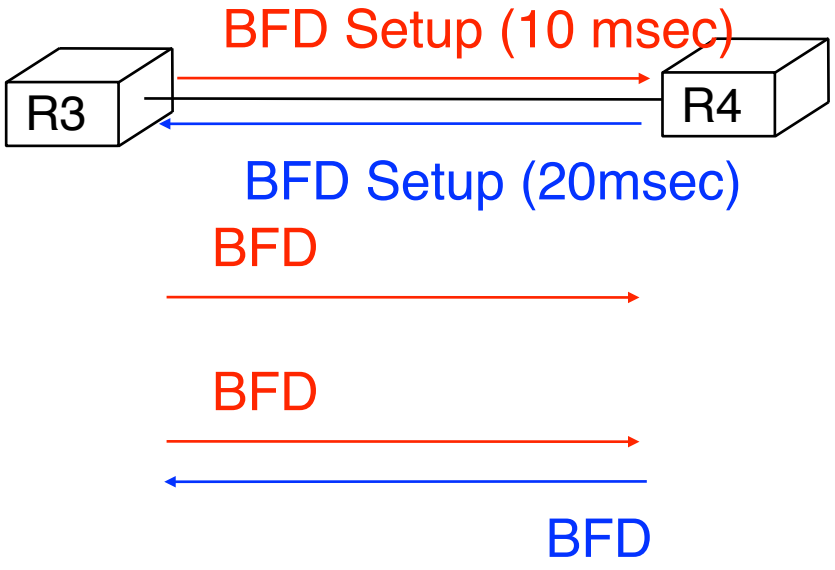
Detecting link/router failures

- Another solution
  - Bidirectional forwarding detection (BFD)
  - Simple UDP based protocol
    - can be implemented on router linecards directly
  - Two modes of operation





⇒ O(10 msec) failure detection becomes possible

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Dealing with link/router failures

- How long does it take to propagate the failure information inside the network ?
  - O(a few hundred msec.) inside a single AS
    - key factors that influence the convergence time are
      - number of prefixes advertised inside IGP
      - time to update one entry inside the IGP
      - measurements indicate an update time for 60 - 100 musec per prefix on high-end router platforms
  - O(tens sec.) across the Internet
    - measurements on the Internet show that a failure is recovered within several tens of seconds, but in some cases it may take several minutes
    - BGP suffers from the path exploration problem that may worsen the convergence time

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MPLS based fault-tolerance

- Can we do better with MPLS ?
  - MPLS forwarding does not depend on routing
    - a MPLS router can forward packets even when routing has not converged provided that a secondary LSP exist
  - Solution
    - Establish secondary LSPs to protect important LSPs
      - secondary LSP is established and maintained inside the network but carries traffic only in case of failure of the primary LSP
    - when an outgoing link or router fails inform headend LSR to stop using the primary LSP and switch all traffic to the protection LSP
      - this operation can be done by the MPLS router itself without any cooperation with other routers provided the protection LSP exists

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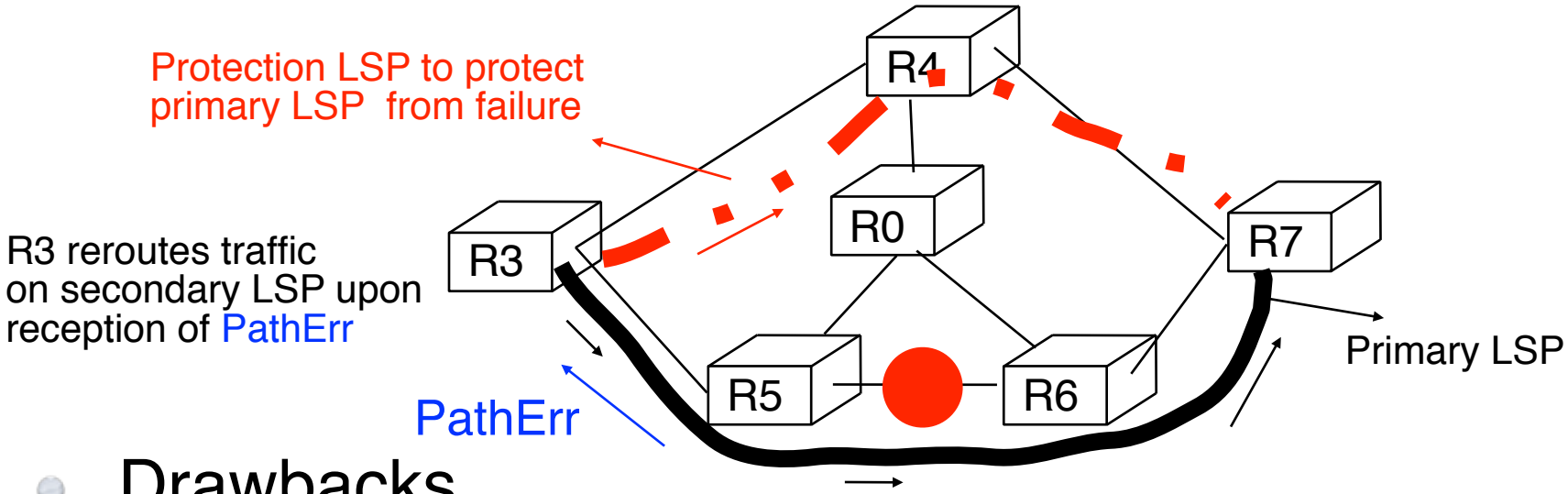
End-to-End secondary LSP

- First solution
  - Secondary LSP established between ingress LSR and egress LSR
  - In case of failure, PathErr message sent to ingress



  - Drawbacks
    - PathErr may take some time to reach ingress LSR
      - packets may be lost between failure detection and reception of PathErr
    - one secondary LSP for each primary LSP

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Selection of path for secondary LSP

- How to select a suitable path for a secondary LSP to protect a primary LSP ?
  - Principle of the solution
    - Ingress LSR selects path for primary LSP by using its path selection algorithm on the entire network topology
    - Knowing the path of primary LSP, ingress LSR computes a path for the secondary LSP by using its path selection algorithm on the network topology where the resources used by primary LSP have been removed
      - secondary LSP must rely on different physical resources than primary LSP
      - this implies that the ingress LSR needs to know the physical resources used by each LSP
        - RSVP carries this information in the RRO object

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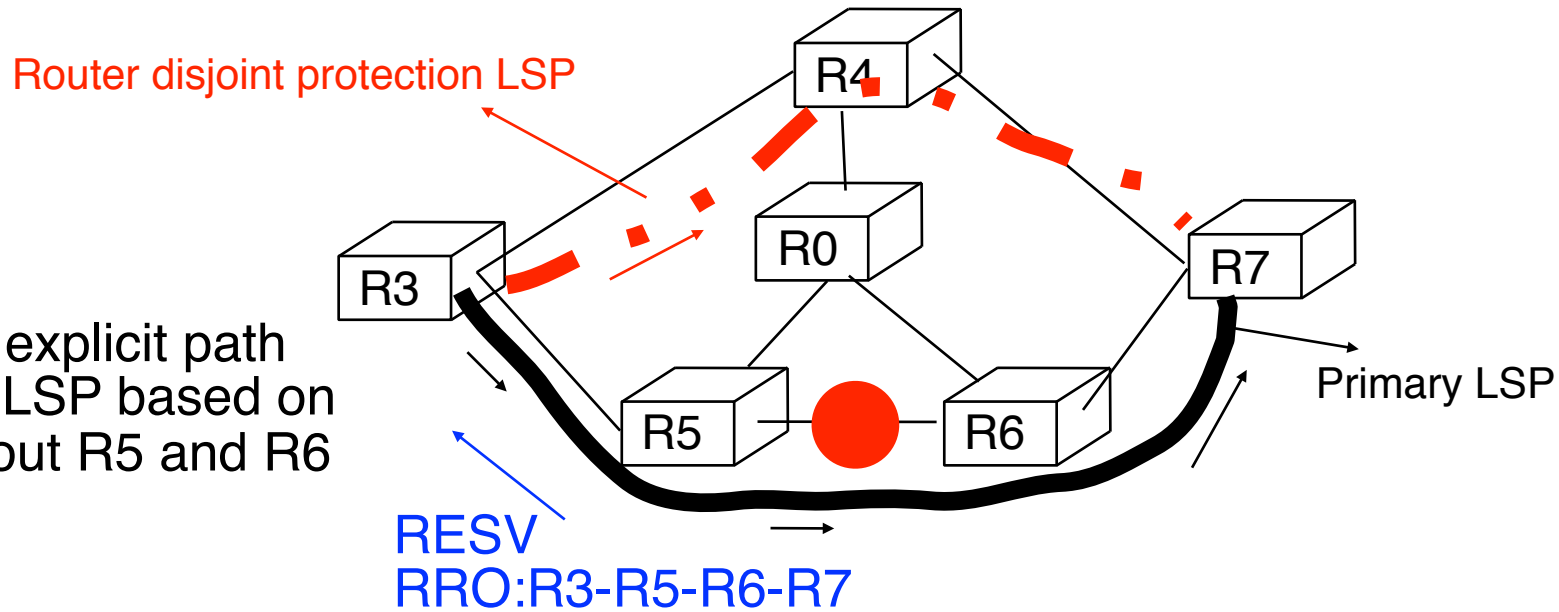
Selection of path for secondary LSP (2)

- Types of end-to-end protection LSPs
  - router-disjoint protection LSP
    - does not utilize the same transit routers as the primary
  - link-disjoint protection LSP
    - does not utilize the same links as the primary LSP



R3 computes explicit path for protection LSP based on topology without R5 and R6

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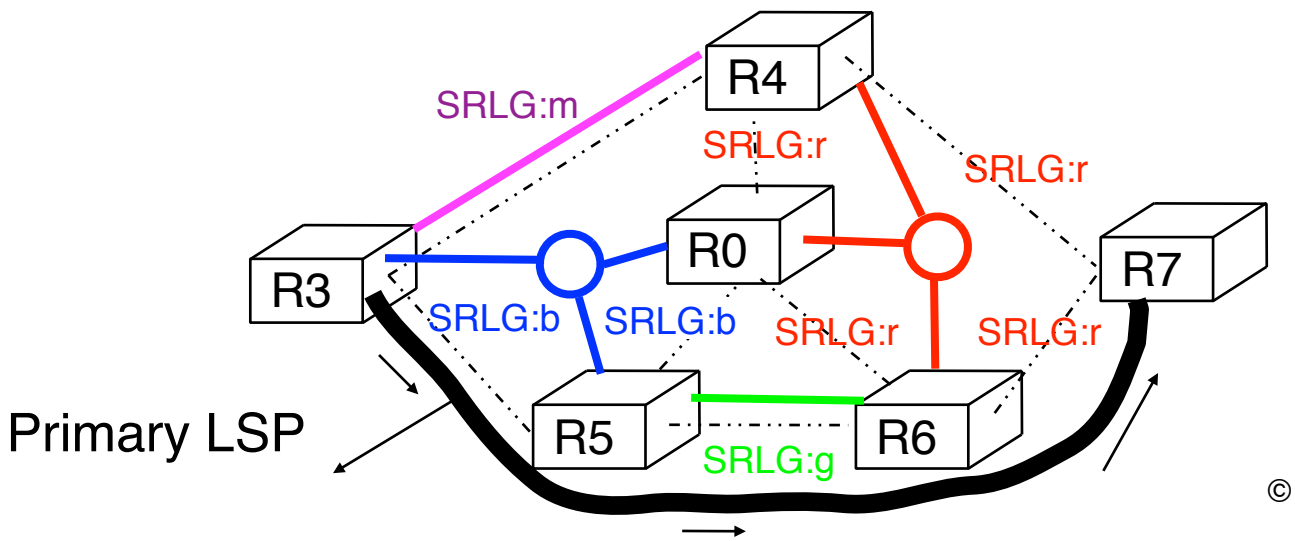
Selection of path for secondary LSP (3)

- Principle of the solution
  - Identify sets of Shared Risk Link Groups
    - a SRLG identifies set of resources that can fail together
      - all SDH circuits going through the same fiber
      - all SDH circuits going through a single TDM switch
      - routers connected to same power source ..
  - Advertise the SRLGs of each resource with IGP
  - Prune links and routers that are part of same SRLG as primary LSP when computing secondary



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Improving MPLS protection

- Issues with end-to-end protection LSPs
  - one protection LSP must be established for each primary LSP
    - doubles the number of LSPs in the network
    - if bandwidth is reserved for primary, bandwidth must be reserved for secondary
  - failure information must travel back to ingress LSR
- It would be interesting to
  - Reduce the time required to switch to the protection LSP
  - Reduce the number of protection LSPs established inside the network

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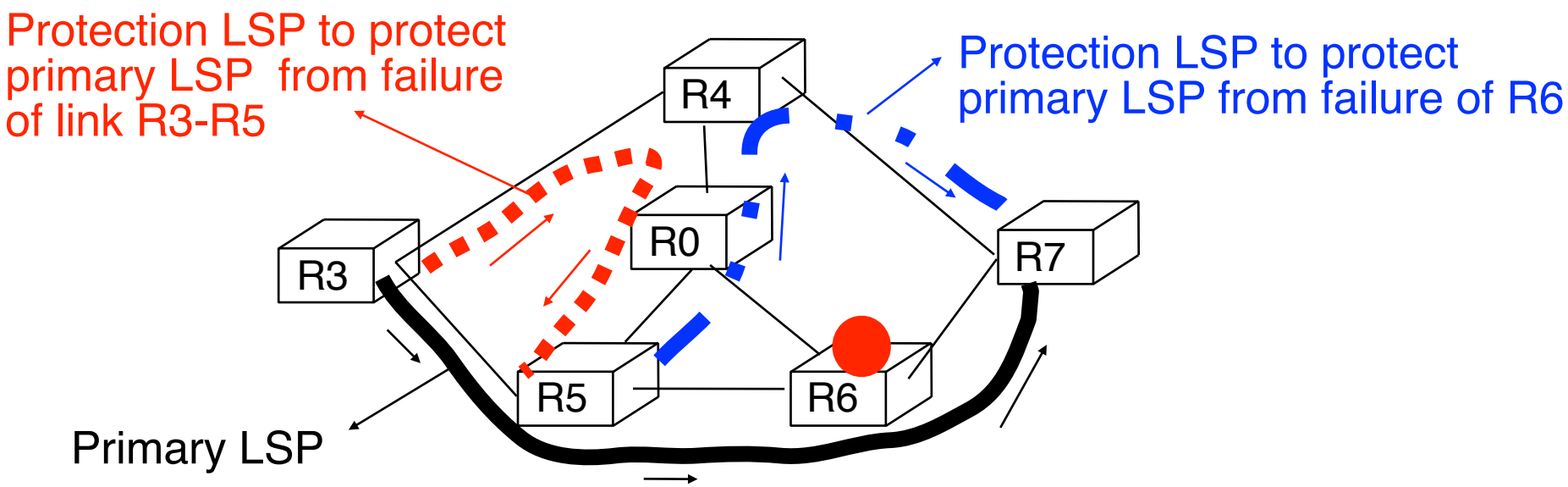
Failures to consider

- Two types of failures
  - Router failure
    - a router and all the links attached to this router fail
  - Link failure
    - a link fails between two adjacent routers



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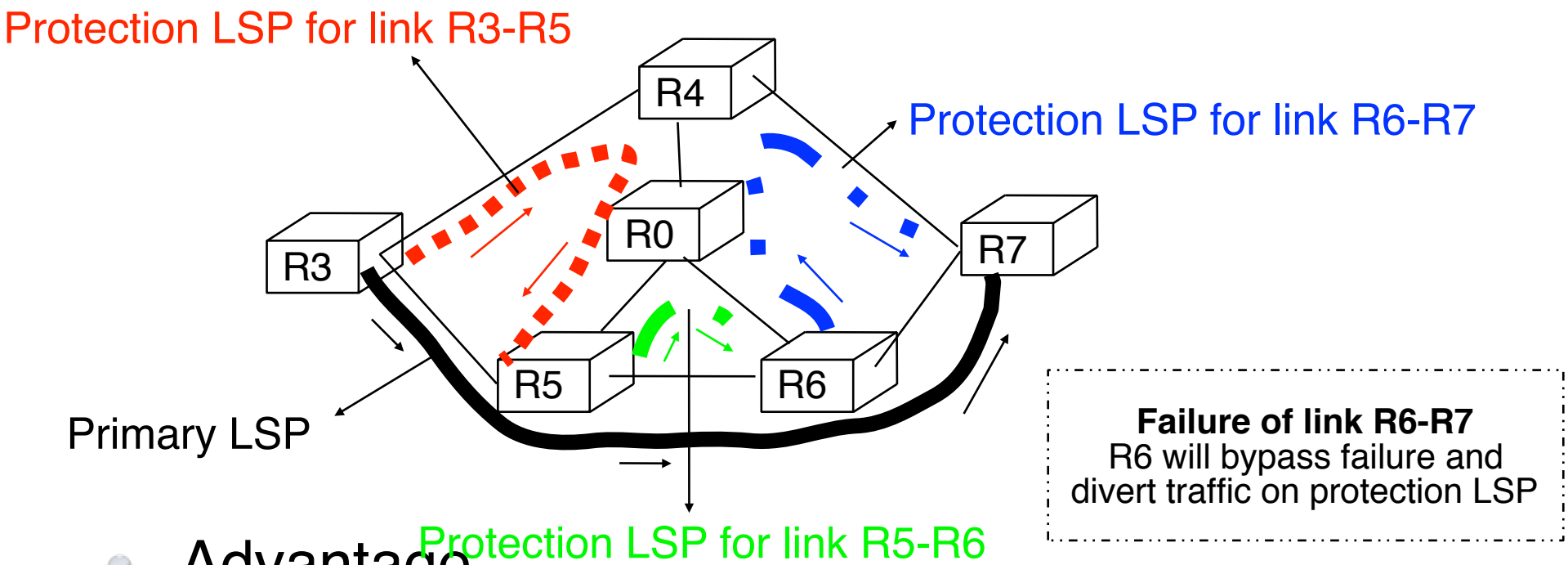
Protecting a complete path

- Principle
  - one protection LSP is used to protect each link



  - Advantage
    - traffic is immediately switched to secondary LSP
  - Drawback
    - a large number of protection LSPs may be required

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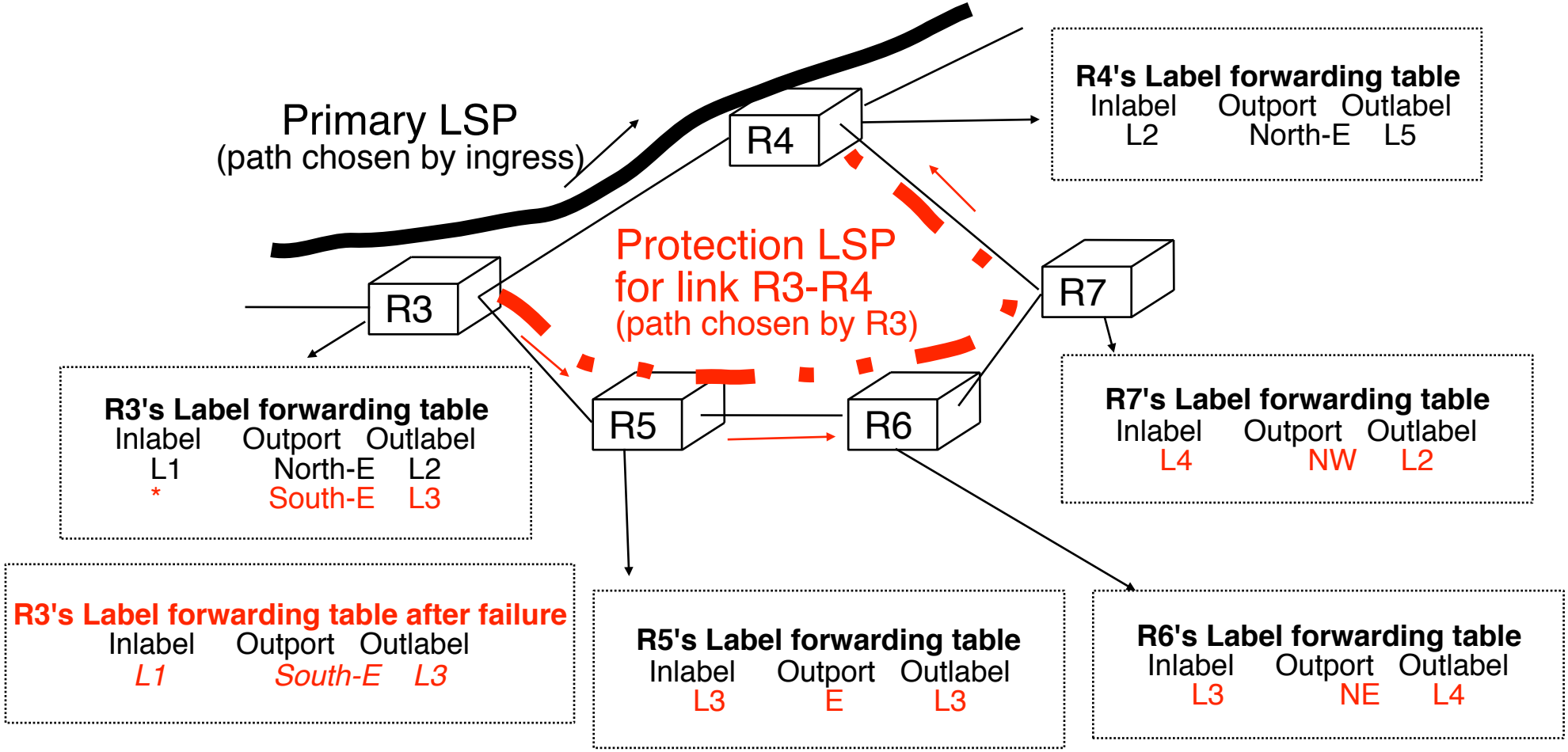
Per-LSP link failure protection

- Principle
  - Protection LSP established by each LSR to protect each link used by primary LSP



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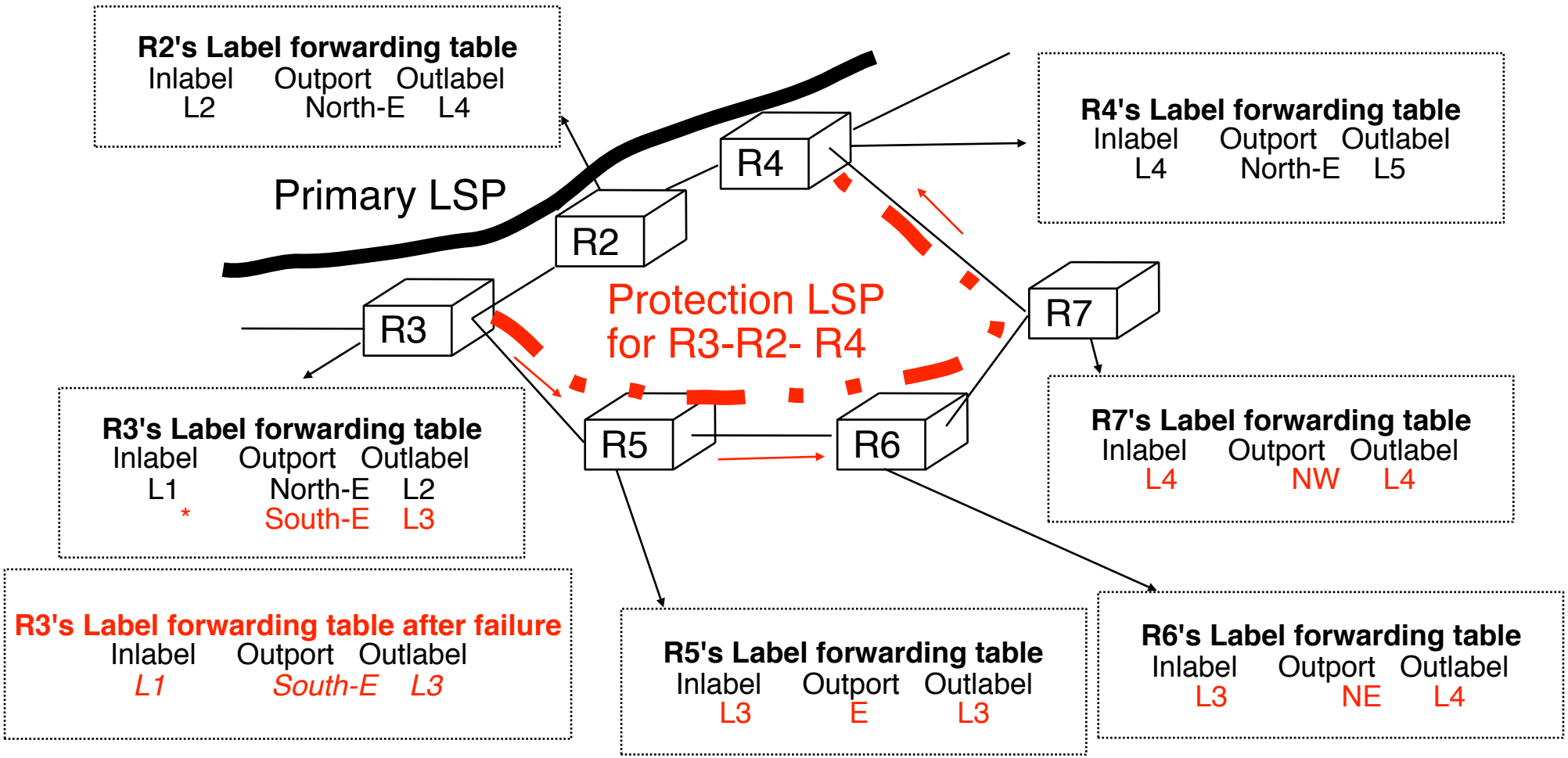
Per-LSP router failure protection

- Principle
  - Protection LSP established by each LSR



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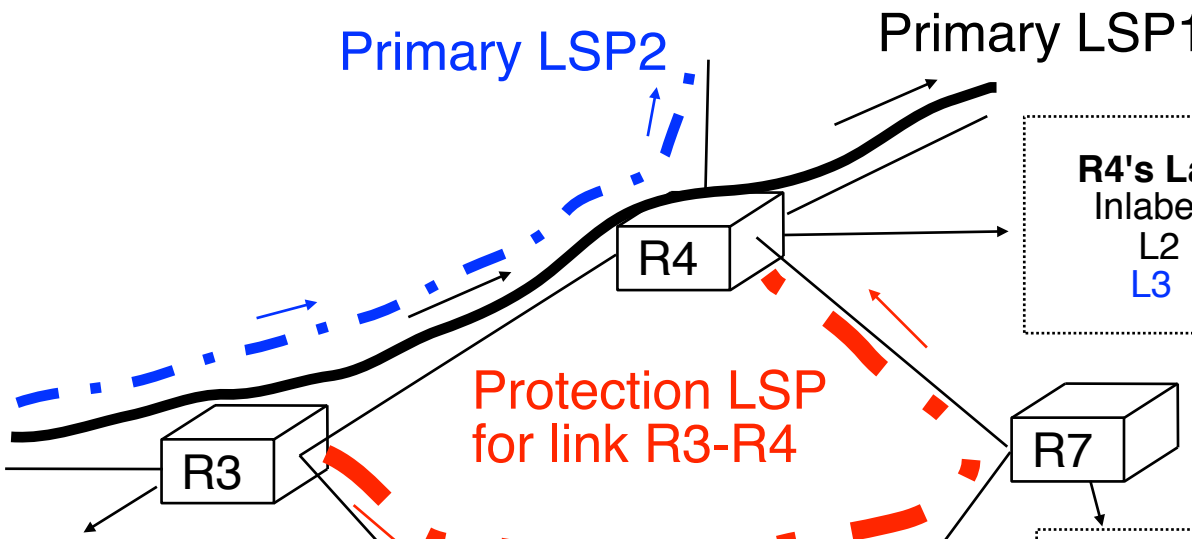
Protecting several LSPs together



R4's Label forwarding table

Inlabel	Outport	Outlabel
L2	North-E	L5
L3	North	L6

R3's Label forwarding table

Inport	Inlabel	Outport	Operation
West	L1	North-E	SWAP(L2)
West	L2	North-E	SWAP(L3)
*	*	South-E	PUSH(L3)

R7's Label forwarding table

Inport	Inlabel	Outport	Operation
SW	L4	NW	POP

R3's Label forwarding table after failure

Inport	*Inlabel*	*Outport*	*Operation*
West	*L1*	*South-E*	*SWAP(L2);PUSH(L3)*
West	*L2*	*South-E*	*SWAP(L3);PUSH(L3)*

R5's Label forwarding table

Inport	Inlabel	Outport	Operation
NW	L3	E	SWAP(L4)

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Dealing with failures

- Dealing with failures
  - Link failures in large IP networks
  - MPLS-based fast restoration
  - IP-based fast restoration

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How to provide sub 50 msec recovery in pure IP networks ?

- First step
  - When a link fails, immediately reroute the packets *at the router that detects the failure to a loop-free alternate router*
    - *This loop-free alternate router is precomputed*
- What is a loop-free alternate router ?
  - A neighbor N is a loop-free alternate that allows S to reach destination D if

$$\text{Distance}(N, D) < \text{Distance}(N, S) + \text{Distance}(S, D)$$

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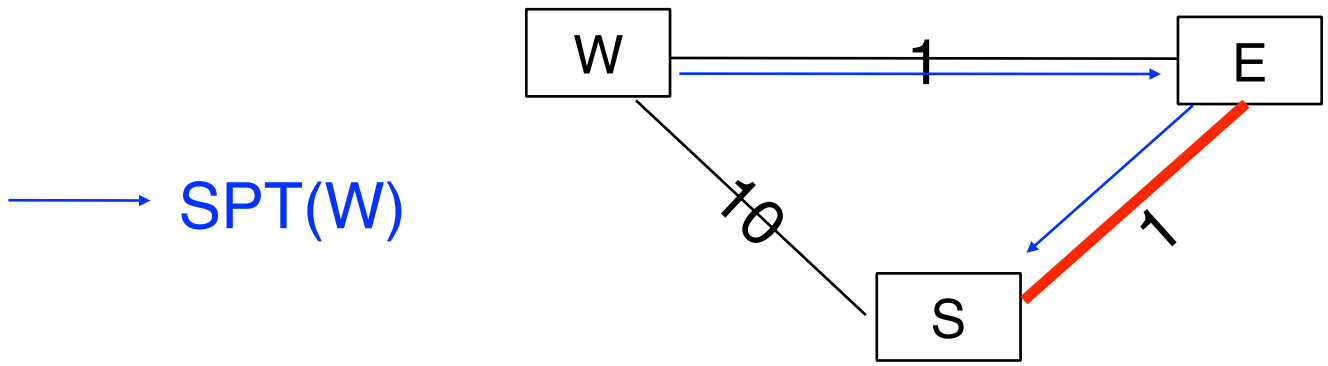
Loop-free alternate



- Loop-free neighbour detection algorithm for protected link S->E
  - For each direct neighbour (S->N_i)
    - Compute SPT(N_i)
    - if (S->E) $\notin$SPT(N_i)
      - then N_i is a candidate loop-free neighbour for all destinations
      - otherwise not

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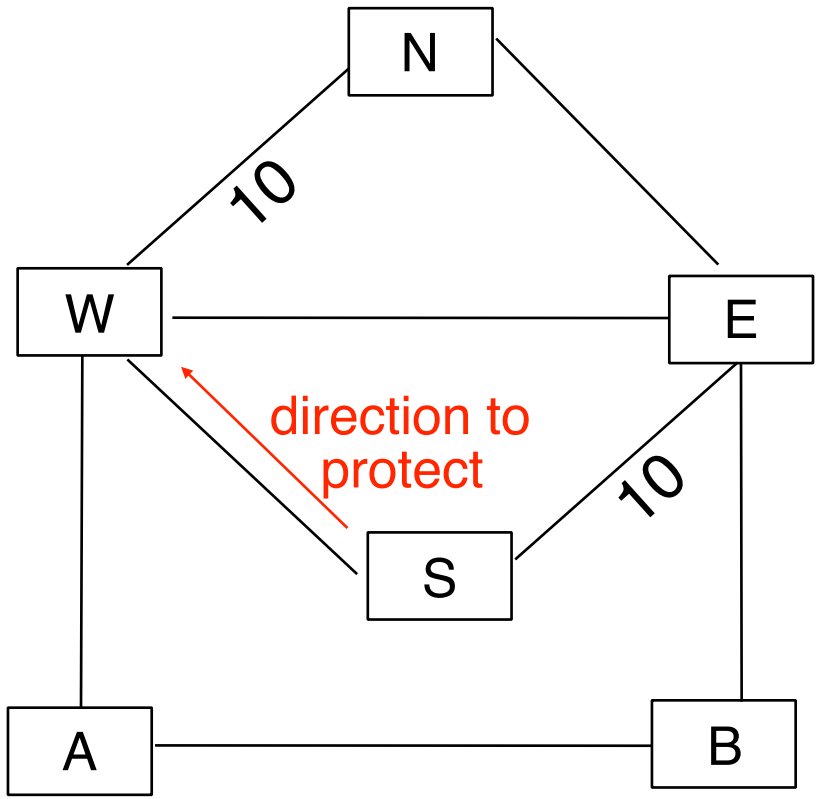
Loop-free alternate

- Example
  - all links have weight=1 except NW and SE

<u>W's routing table</u>
N : East via E
S : SouthEast
E : East
A : South
B : South via A
East via E



<u>E's routing table</u>
N : NothWest
S : SouthWest
W: West
B : South
A : South via B
West via W

<u>S's routing table</u>
All : via W

- If S->W fails, E is a loop-free neighbor
  - all S->W's packets sent to E will not loop

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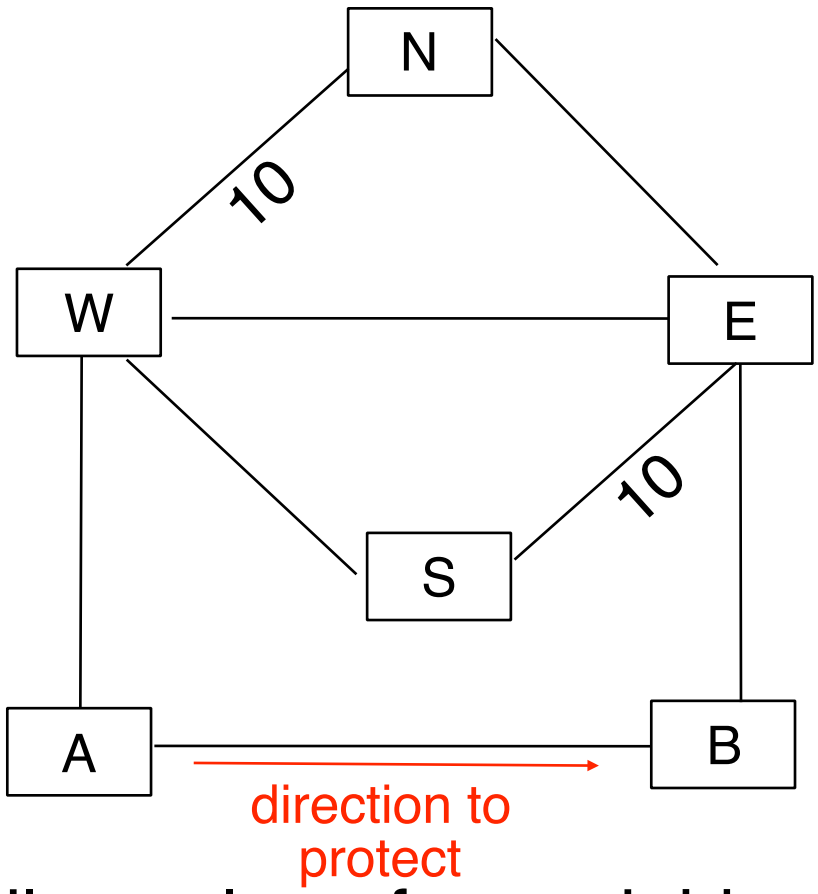
Loop-free alternate (2)

W's routing table
N : East via E
S : SouthEast
E : East
A : South
B : South via A
East via E

A's routing table
N,S : North via W
W : North
B : East
E : North via W
East via B



B's routing table
E : North
A : West
N,S : North via E
W : West via A
North via E

- If A->B fails, no loop-free neighbour for destination B
  - W would return to A the packets towards node B

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Evaluation of loop-free alternates

- Question
  - How many links can be quickly protected by immediately switching over all the traffic that they carry to a loop-free neighbour ?

Network	Links	LFA
Abilene	28	42%
GEANT	72	66%
ISP1	114	54%
ISP2	26	15%
ISP3	265	65%

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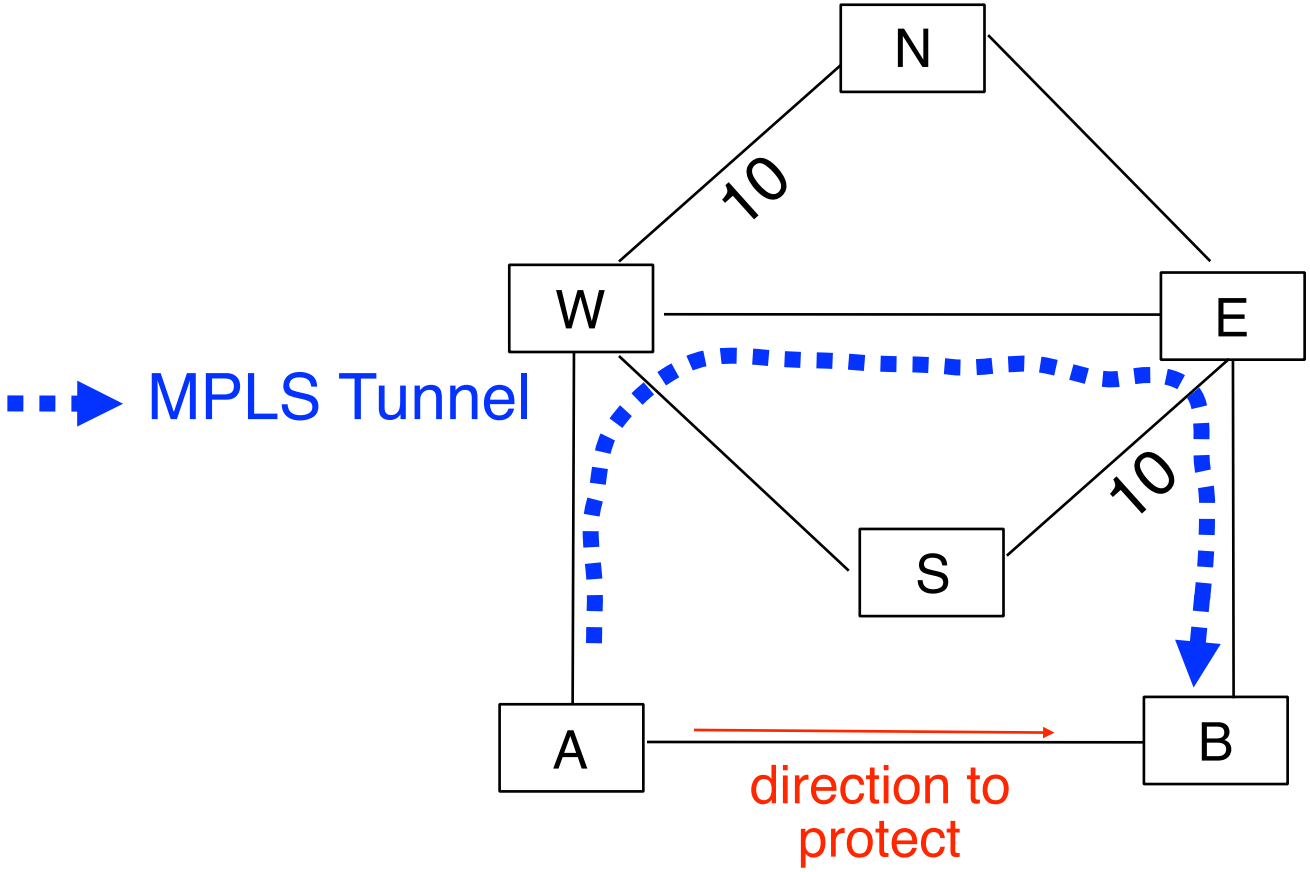
Loop-free alternate routers

- How to improve the coverage ?
  - Use non-neighbours as alternate routers
  - Simple solution
    - MPLS tunnel to protect failed link



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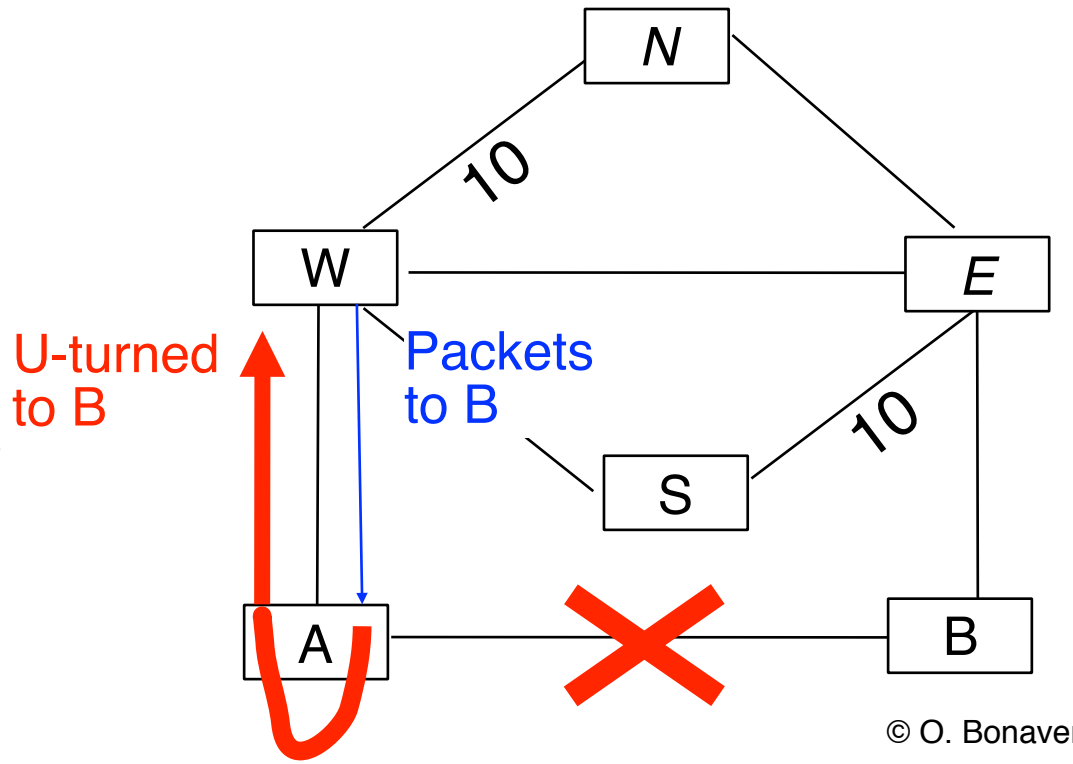
U-turns

- Principle
  - If there is no loop-free neighbour, a neighbour of our neighbours might be loop-free...
  - When failure occurs, return the packets to sender who will send them to its loop-free neighbour
    - *Assumes hardware support on interfaces*

<u>W's routing table</u>
N : East via E
S : SouthEast
E : East
A : South
B : South via A
East via E
Loop-free neighbours
If W->A fails
Use E to reach B



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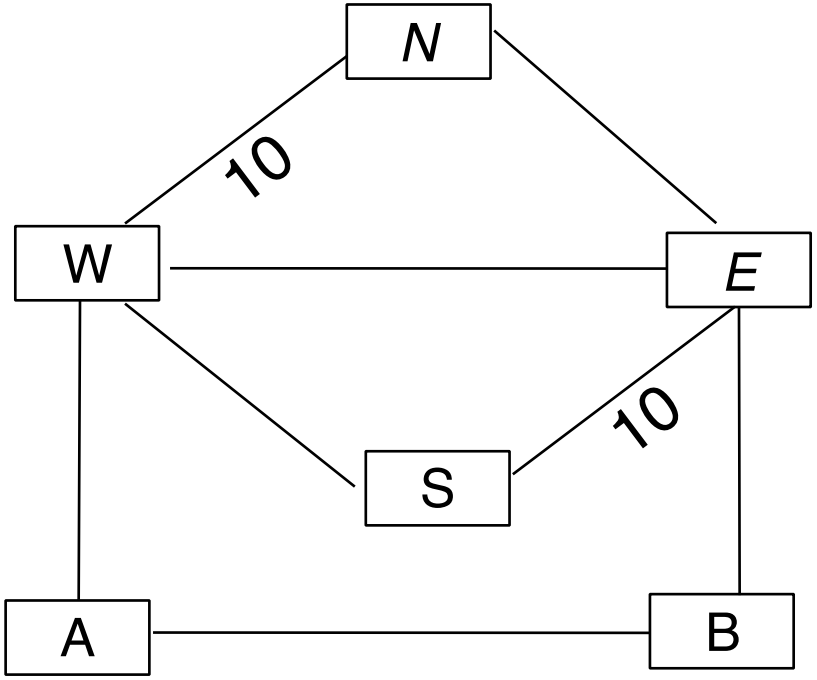
Not via addresses

- Assign special not via addresses to each node
  - BnvA : reach B without passing via A
  - BnvE : reach B without passing via E



<u>N's routing table</u>
All : via E

<u>E's routing table</u>
N : NorthWest
S : SouthWest
W: West
B, BnvA : South
A : South via B
West via W
BnvE : West

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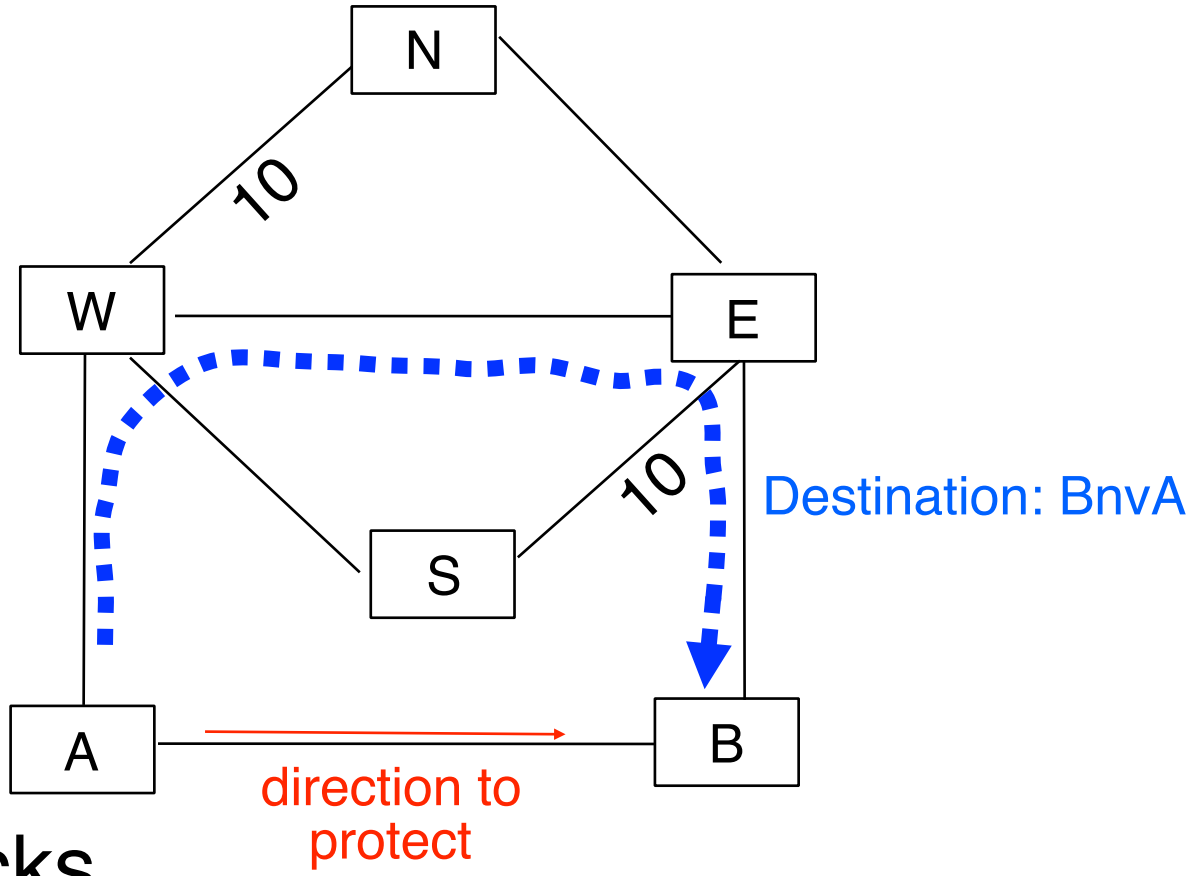
Usage of not-via-addresses

- Used as destination addresses for protection tunnels



- Drawbacks
  - Increases number of addresses
  - Increases number of SPF computations

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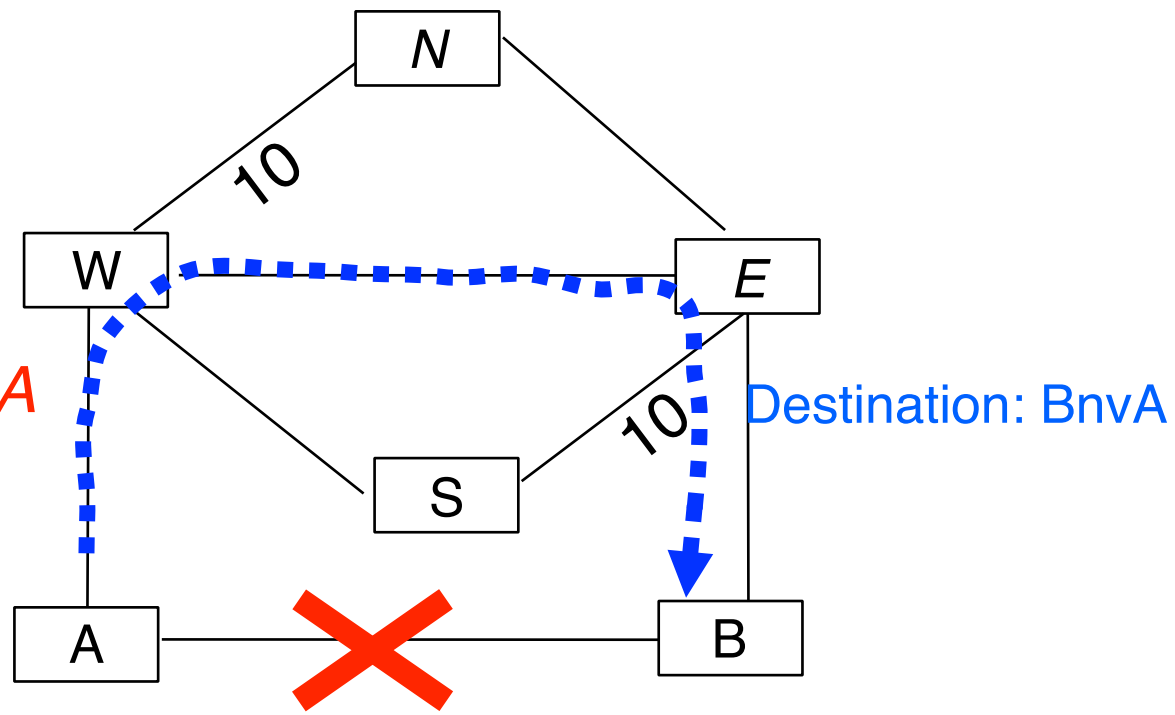
Are IP protections sufficient ?

- Consider the failure of link A->B
  - A immediately updates its FIB to use tunnel

A's routing table
N,S : North via W
W : North
~~B : East~~
E : North via W
~~East via B~~
B : via tunnel to BnvA



  - Is this sufficient to avoid all packet losses ?

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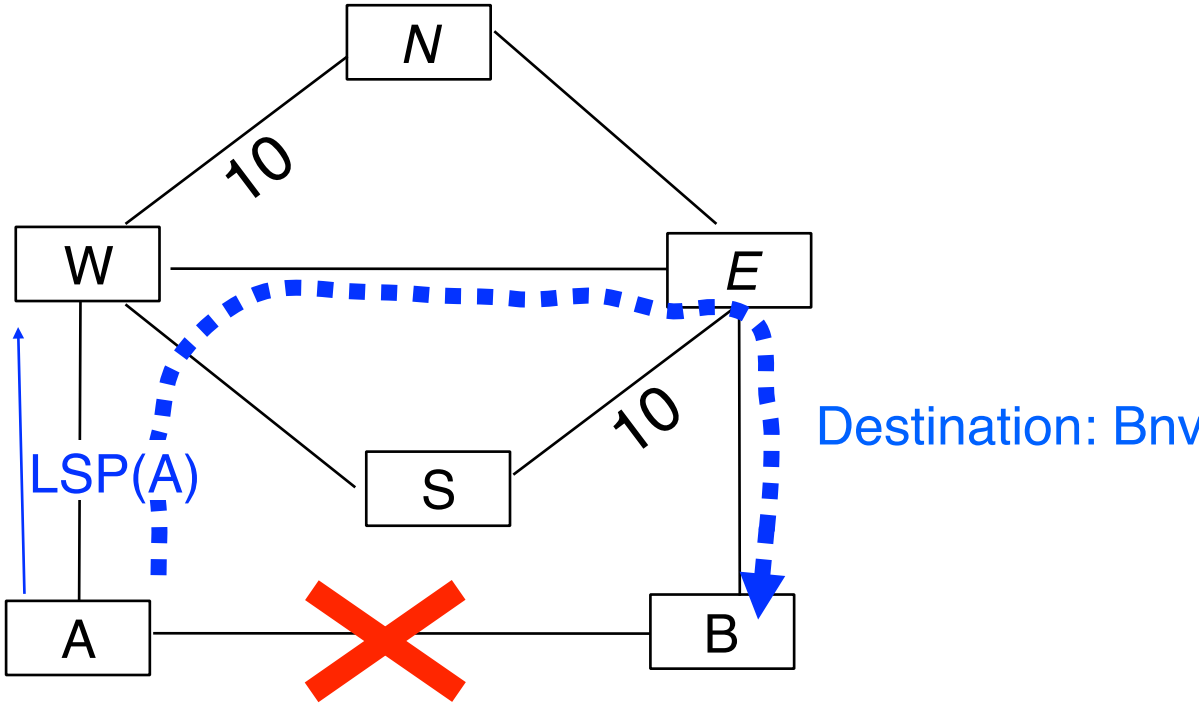
Are IP protections sufficient ? (2)

- Unfortunately, the protection tunnel is not optimal
  - A will flood its new link-state packet and all routers will eventually update their FIB

A's routing table
N,S : North via W
W : North
~~B : East~~
E : North via W
~~East via B~~
B : via tunnel to BnvA



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Are IP protections sufficient ? (3)

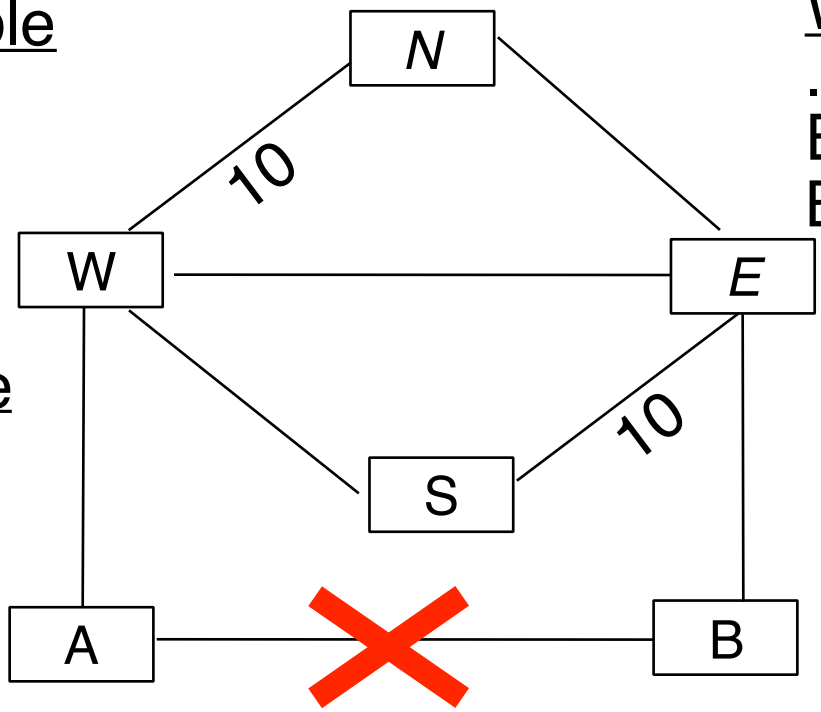
- W updates its FIB before A
  - Everything is fine, no packets are lost
  - A updates its FIB before W
    - Packets towards B loop between A and W

A's new routing table

...
E : North via W
B : North via W

A's old routing table

...
E : North via W
B : via tunnel to E



W's old routing table

...
E : East
B : South via A
East via W

W's new routing table

...
E : East
B : East via E

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How to avoid transient loops during FIB updates ?

- Two solutions are possible
  - Distributed ordering of the updates of the FIBs
    - could be included in OSPF/IS-IS
  - Metric increments to avoid transient loops
    - can also handle all non-urgent topology changes
      - link brought up/down for maintenance
      - router reboot
      - change in link weights
- As of today, none of these techniques is being standardised

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